

Chapter 13: Nuclei

Formula Sheet + Tips & Tricks

Formula Sheet

Standard Constants used in this Chapter:

$$m_p = 1.00728 \text{ u} \quad m_n = 1.00867 \text{ u} \quad m_e = 0.00055 \text{ u}$$

$$h = 6.6 \times 10^{-34} \text{ J-s} \quad c = 3.0 \times 10^8 \text{ m s}^{-1}$$

$$1 \text{ MeV} = 1.60 \times 10^{-13} \text{ J} \quad 1 \text{ u} = 931.5 \text{ MeV (i.e. } 1 \text{ u} \times c^2 = 931.5 \text{ MeV)}$$

I. Mass-Energy Relation

Mass-Energy Equivalence

$$E = mc^2$$

E = energy released/absorbed (J); m = mass converted (kg); c = speed of light

Percentage Change in Mass

$$\% \text{ increase} = \frac{\Delta m}{M} \times 100$$

Δm = mass gained/lost; M = original mass of body

II. Pair-Production and Annihilation of Matter

Minimum Energy for Pair Production

$$E_{min} = 2m_e c^2$$

m_e = rest mass of electron (equal to rest mass of positron)

Photon Energy-Wavelength Relation

$$E = \frac{hc}{\lambda}$$

h = Planck's constant; λ = wavelength of the photon

Particle-Antiparticle Annihilation Energy

$$E = 2mc^2$$

m = rest mass of each particle (e.g. proton and antiproton)

III. Mass Defect and Binding Energy**Mass Defect**

$$\Delta m = [Zm_p + (A - Z)m_n] - M_{nucleus}$$

Z = atomic number; A = mass number; $M_{nucleus}$ = actual nuclear mass

Binding Energy

$$BE = \Delta m \text{ (in u)} \times 931.5 \text{ MeV}$$

Δm expressed in atomic mass units (u)

Binding Energy per Nucleon

$$\frac{BE}{A}$$

Measure of nuclear stability; higher value $\Rightarrow$ more stable nucleus

IV. Q-Value of Nuclear Reactions**Q-Value**

$$Q = (\Sigma m_{reactants} - \Sigma m_{products}) \times 931.5 \text{ MeV}$$

Positive Q = energy released (exothermic); Negative Q = energy absorbed

V. Nuclear Fission**Fission Energy**

$$E = \Delta mc^2$$

Δm = mass defect in the fission process

Energy-kWh Conversion

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

Used to express large nuclear energies in electrical units

Fission Rate

$$\frac{dN}{dt} = \frac{P}{E_{fission}}$$

P = power output (W); $E_{fission}$ = energy released per fission (J)

VI. Nuclear Fusion

Fusion Energy

$$E = \Delta mc^2$$

Δm = mass defect between fusing nuclei and product nucleus

Useful Energy with Efficiency

$$E_{\text{useful}} = \eta \times E_{\text{total}}$$

η = fractional efficiency of the generator/reactor

Tips, Tricks & Hacks

- Mass defect $\Delta m = [Zm_p + (A - Z)m_n] - M_{\text{nucleus}}$ is always positive for a stable nucleus — this “missing mass” converted to energy is precisely what binds the nucleus together.
- Binding energy per nucleon peaks around mass number $A \approx 56$ (iron region) — nuclei lighter than this release energy on *fusion* (moving right, up the curve), nuclei heavier release energy on *fission* (splitting to move up the curve too); both processes move nuclei toward the peak.
- $1 \text{ u} = 931.5 \text{ MeV}/c^2$ — memorise this conversion factor, it’s used in virtually every mass-defect/Q-value numerical.
- Q-value of a reaction = (mass of reactants – mass of products) c^2 — positive Q means energy is released (exothermic, reaction proceeds spontaneously), negative Q means energy must be supplied.
- Radioactive decay is a random, spontaneous process independent of external conditions (pressure, temperature) — a common conceptual trap is expecting decay rate to change with the environment; it doesn’t.
- Half-life and mean life are related by $T_{1/2} = 0.693 \tau$ — mean life is always *longer* than half-life; don’t accidentally swap them in a numerical.
- Activity $A = \lambda N = A_0 e^{-\lambda t}$ decays with the *same* exponential law as N itself — so activity halves every half-life too, just like the number of undecayed nuclei.
- Pair production needs a photon of energy $\geq 2m_e c^2 = 1.02 \text{ MeV}$ (creates an electron-positron pair); annihilation is the reverse, producing (usually two) photons of 0.51 MeV each — a nice paired fact to remember together.